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Effects of Temperature on Specific Daily Metabolic Demand and Growth Scope of Sub-Adult and Adult Smallmouth Bass

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ABSTRACT

Specific daily metabolic demand (MET) and growth scope were determined for sub-adult (older juvenile) and adult smallmouth bass over the temperature range of 18-30 °C. MET generally increased with increasing temperature and decreased with increasing fish weight, following patterns typically observed for fishes. Growth scope for sub-adult and adult smallmouth bass reached a maximum at 22 °C and declined rapidly at higher temperatures to just 20% of maximum at 30 °C. Sustained periods in which water temperatures substantially exceed 22 °C are expected to lead to restrictions in growth scope for older juvenile and adult smallmouth bass, resulting in diminished growth rates, size, and fecundity.

INTRODUCTION

Smallmouth bass (*Micropterus dolomieu*) is a popular sport fish and an important ecological component of Ozark streams. During the last 50 years, declines in smallmouth bass abundance have been documented in many streams in the northern Ozarks and Ozark border region of Missouri (Bauman 1945, Pflieger 1975, Fajen 1981). In some streams, replacement of smallmouth bass by largemouth bass (*Micropterus salmoides*) has been noted (Fajen 1981, Sowa and Rabeni 1995). In Ozark border streams, higher summer water temperatures and greater relative areas of pools are associated with relatively low smallmouth bass densities and relatively high largemouth bass densities; these changes in stream habitat are thought to have resulted from alterations in land use (Sowa and Rabeni 1995).

Evidence from a recent study suggests that interspecific differences in maximum daily consumption ($C_{\max}$) may partly explain declines in smallmouth bass populations and increases in largemouth bass abundance in streams with elevated summer temperatures (Zweifel et al. 1999). Over a fish's thermal tolerance range, $C_{\max}$ initially increases with temperature, peaks or plateaus at some intermediate temperature ($T_{\max C}$), and then declines rapidly thereafter (Kitchell et al. 1977, Jobling 1993). For sub-adult and adult smallmouth bass, $C_{\max}$ begins to decline at 22 °C, while $C_{\max}$ does not begin to decline until 26 °C for largemouth bass of similar sizes and ages (Zweifel et al. 1999). As a result of sharp declines in $C_{\max}$ coupled with increasing metabolic costs at temperatures above $T_{\max C}$, a fish's growth scope (the difference between $C_{\max}$ and energy costs;

Warren and Davis 1967) drops precipitously at temperatures above $T_{\max C}$, and an individual's ability to grow becomes increasingly limited irrespective of food availability (Jobling 1997). Thus, streams in which summer temperatures substantially exceed 22 °C are generally expected to be unfavorable for growth of adult smallmouth bass and could lead to declines in their condition, size, fecundity, and ultimately, population density. However, temperatures up to 26 °C would not be expected to negatively affect adult largemouth bass growth (Zweifel et al. 1999).

Although relations between water temperature and $C_{\max}$ have been established for sub-adult and adult smallmouth bass (Zweifel et al. 1999), no information is currently available regarding the influence of temperature on their energy costs. A bioenergetics model has been developed for smallmouth bass (Shuter and Post 1990), but this model was constructed specifically for and using data gathered from age-0 fish only. Roell and Orth (1993) developed a model to assess trophic basis of smallmouth bass production in Virginia streams, but their model primarily used parameters borrowed from other species, primarily largemouth bass. Since relations between water temperature and metabolism frequently differ among fish species (Hewett and Johnson 1992, Jobling 1997) and between sizes and ages of a particular species (Post 1990, Hartman and Brandt 1995), existing smallmouth bass bioenergetics models may not accurately describe rate-temperature relations for adult smallmouth bass metabolism. Knowledge of how temperature affects energy costs for fishes is needed in addition to $C_{\max}$ data to accurately define the influence of temperature on growth scope and to determine thermal limits for growth of a species. The objectives of this study were to determine how water temperature affects specific daily metabolic demand and growth scope of sub-adult and adult smallmouth bass over the temperature range of 18-30 °C.

METHODS AND MATERIALS

Smallmouth bass were collected by electrofishing from northern Ozark streams during 1999 and 2000. Two size classes of fish were collected: small (50-100 g wet wt., sub-adult (older juvenile) fish) and large (150-280 g wet wt., adult fish). These size classes encompass the range of fish sizes used by Zweifel et al. (1999). Fish were transported to the laboratory and acclimated to laboratory conditions over 14 d. Fish were held individually in perforated 9.5 L plexiglas chambers submerged in two adjacent 1135 L tanks equipped with water recirculation, biofiltration, and temperature control systems. Acclimation conditions included a 16 h light : 8 h dark photoperiod and a water temperature of 22 °C $\pm$ 0.5 °C. Fish were not fed during this acclimation period, and tanks were covered with plastic sheeting to minimize contact with humans and reduce fish stress.

Specific daily metabolic demand (MET) was determined for four smallmouth bass from each size class at each of four temperatures (18, 22, 26 and 30 °C $\pm$ 0.5 °C). Prior to testing at 18, 26, and 30 °C, water temperature was changed at a rate of 1-2 °C d⁻¹ until the test temperature was reached. Fish were acclimated to test temperatures for 14 d before each experiment. Throughout the

acclimation periods, fish were fed fathead minnows (*Pimephales promelas*) daily at approximately 50% of maximum ration (Zweifel et al. 1999). Fish were starved for 24 h immediately prior to testing at each temperature. During experiments, four fish from the same size class were placed individually into four flow-through 13.24 L vinyl metabolic chambers (Cech 1990) submerged in adjacent 1135 L tanks. Fish were held in metabolic chambers for 48 h and dissolved oxygen concentrations (nearest 0.1 mg L⁻¹) were recorded simultaneously for water entering and exiting each chamber every five minutes using a PT4 monitor (Point Four Systems, Inc.). Oxygen uptake by each fish was calculated as:

$$(IO_2 - FO_2) * D$$

where IO_2 and FO_2 are dissolved oxygen concentrations (mg L⁻¹) of the inflow and outflow water, respectively, and D is the rate of water flow through each chamber (L h⁻¹). Water flow rate through chambers was regulated so that dissolved O_2 concentrations were depleted by 1-1.5 mg L⁻¹ during experiments; dissolved O_2 concentrations in outflowing water were always ≥ 5.7 mg L⁻¹. Immediately following testing at each temperature, fish were weighed to the nearest 0.1 g. Specific daily oxygen demand (g O_2 g⁻¹ d⁻¹) was calculated for each fish by dividing mean oxygen uptake over the latter 24 h of each experiment (mean of 288 readings of g O_2 h⁻¹) by fish weight and multiplying by 24 h. Data from only the latter 24 h of each experiment were used because previous research has indicated that oxygen uptake by fishes can be erratic during the first 24 h of respirometry experiments but stabilizes thereafter (Hartman and Brandt 1995, Zweifel 2000). Specific daily oxygen demand was then converted into prey-equivalent units (g prey g⁻¹ d⁻¹) to enable direct comparison with existing C_{max} data. First, specific daily oxygen demand was converted to specific daily caloric demand using an oxy-caloric multiplier of 3,280 cal g O_2 ⁻¹ (Adams and Breck 1990). Specific daily caloric demand was then converted into specific daily metabolic demand (MET, g g⁻¹ d⁻¹) by dividing by the caloric density of fathead minnows (921 cal g⁻¹ wet wt., Hayward, unpublished data). Each fish was used only once during experiments.

Analysis of variance (ANOVA) followed by the least significant differences (LSD) procedure for separation of means were used to test for differences in MET within each size class over the four temperatures. Data were log-transformed prior to ANOVA to correct for unequal variances.

Direct comparison of MET data from this study with C_{max} data for sub-adult and adult smallmouth bass (Zweifel et al. 1999) required that MET values be estimated for a 114 g fish, as C_{max} data from Zweifel et al. (1999) were standardized to fish of that body weight. To estimate MET for a 114 g smallmouth bass at 18, 22, 26, and 30 °C, power functions were fit to log-transformed responses of MET to fish weight at each temperature using proc GLM in SAS. The linear form of these functions were:

$$\ln(MET) = b * \ln(W) + a$$

where W is fish weight (g) and a and b are regression constants. The resulting power function models were used to estimate MET for a 114 g fish at each temperature.

Curves portraying growth scope of smallmouth bass over the temperature range of 18-30 °C were developed for a 114 g fish. First, mean values of C_{max} at

18, 22, 26, and 30 °C (Zweifel et al. 1999) were used to calculate energy losses from egestion (F), excretion (U), and specific dynamic action (SDA). Values for F, U, and SDA (g/g/d) were calculated at each temperature as:

$$F = 0.104 * C_{\max}$$

$$U = 0.068 * (C_{\max} - F)$$

and

$$SDA = 0.16 * (C_{\max} - F)$$

where $C_{\max}$ is the mean value for maximum daily consumption at each temperature (Zweifel et al. 1999), and constants in each equation are values given for centrarchids by Hewett and Johnson (1992). Estimates of energy losses due to F, U, and SDA were then added to mean values for MET at each temperature to generate a curve representing energy losses from these four sources combined ($\text{g g}^{-1} \text{d}^{-1}$). The difference between curves relating $C_{\max}$ and energy losses from MET, F, U, and SDA combined represented growth scope at each temperature.

RESULTS

For adult fish (mean wt. 202 g), specific daily prey demand (MET) ranged from about 1% to 1.6% body wt. d^{-1} and increased significantly from 18 to 22 °C ($p < 0.05$), but did not change significantly with increasing temperature from 22 to 30 °C ($p > 0.05$; Figure 1). For sub-adult (older juvenile) fish (mean wt. 71 g), MET ranged from about 2.1% to 3.6% body wt. d^{-1} and did not increase significantly from 18 to 26 °C ($p > 0.05$), but MET was significantly higher at 30 °C than at 18 and 22 °C ($p < 0.05$) and was marginally higher at 30 °C than at 26 °C ($0.1 > p > 0.05$).

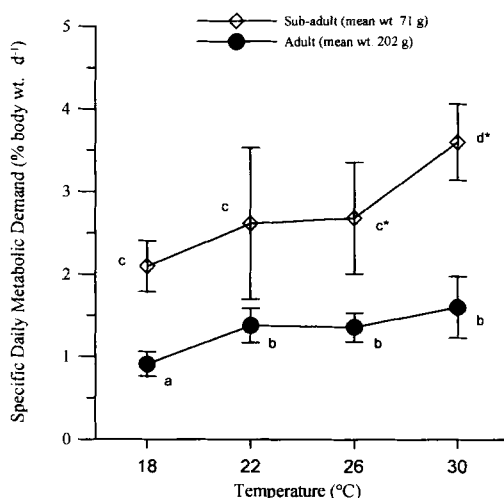


Figure 1. Specific daily metabolic demand for two size classes of smallmouth bass at 18, 22, 26, and 30 °C. Values shown are means and 95% confidence intervals. $n=4$ for each data point. Within each size class, means with the same letter are not significantly different ($p > 0.05$). Values marked with asterisks were marginally different ($p = 0.08$).

Power functions provided good fits to responses of MET to fish weight at all four temperatures ($r^2 = 0.82, 0.73, 0.67$, and 0.97 at $18, 22, 26$, and $30\text{ }^{\circ}\text{C}$, respectively; $p < 0.05$ at all temperatures; Figure 2). Predicted values of MET for a 114 g smallmouth bass using these power function models were 1.51% , 1.96% , 1.73% , and 2.42% body wt. d^{-1} at $18, 22, 26$, and $30\text{ }^{\circ}\text{C}$, respectively.

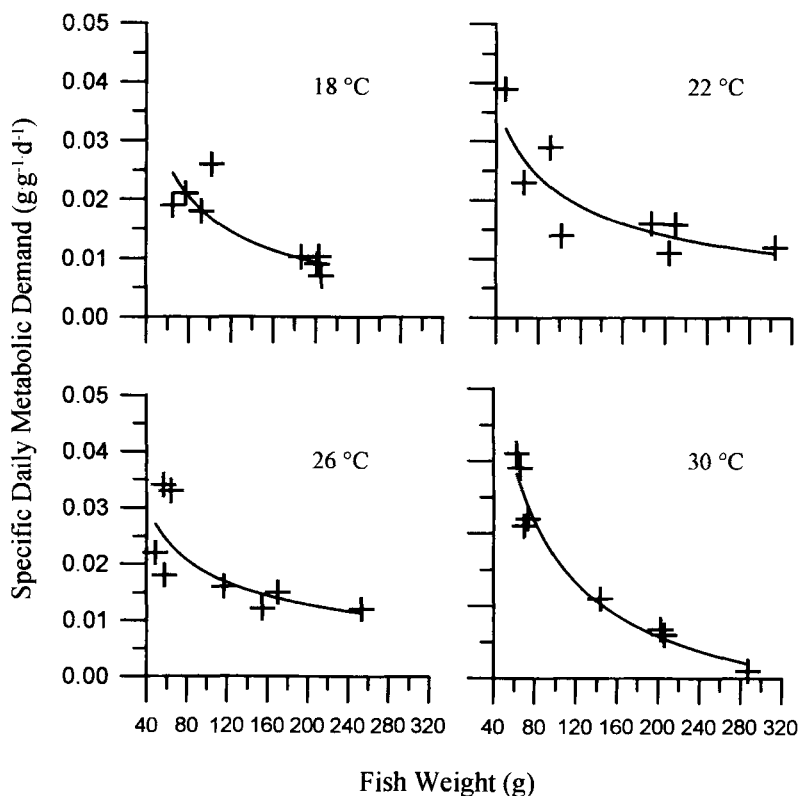


Figure 2. Specific daily metabolic demand in relation to fish weight for smallmouth bass at $18, 22, 26$, and $30\text{ }^{\circ}\text{C}$. Curves shown are power functions fit to data.

Growth scope for a 114 g smallmouth bass increased from 18 to $22\text{ }^{\circ}\text{C}$ and declined at higher temperatures (Figure 3). Changes in C_{max} with temperature were more dynamic than those of functions describing energy losses and were most responsible for changes in growth scope from 18 to $30\text{ }^{\circ}\text{C}$. Growth scope was always positive in the temperature range of 18 - $30\text{ }^{\circ}\text{C}$ but declined rapidly above optimum growth temperature ($22\text{ }^{\circ}\text{C}$) to 62% , 51% , 41% , 30% , and 20% of maximum growth scope at temperatures of $26, 27, 28, 29$, and $30\text{ }^{\circ}\text{C}$, respectively. Energy losses due to resting metabolism, egestion, excretion, and specific dynamic action combined ranged from 55% to 84% of C_{max} over the temperature range of 18 - $30\text{ }^{\circ}\text{C}$ and increased in proportion to C_{max} with increasing temperature (Figure 3).

DISCUSSION

Specific daily metabolic demand (MET) for smallmouth bass generally increased with increasing temperature, although relations between MET and temperature were not exponential as is typically observed for fishes (Kitchell et al. 1977, Brett 1979, Jobling 1993). However, many exceptions to the general pattern of exponential increases in metabolism with increasing temperature have been reported, including instances in which metabolism becomes independent of temperature, usually in the middle of a species' viable temperature range (Cossins and Bowler 1987, Jobling 1993). This phenomenon was observed during this study, as there were no significant changes in MET between 22 °C and 26 °C for either size class of smallmouth bass.

Specific daily metabolic demand for smallmouth bass decreased with increasing fish size, following the pattern typically observed for fishes (Jobling 1993). Estimates of slopes of functions relating MET to fish weight for smallmouth bass at each of the four temperatures ranged from 0.53 to 0.86, similar to the range of 0.65 to 0.9 within which most reported estimates of slopes of relations between fish weight and metabolic rate lie (Jobling 1993).

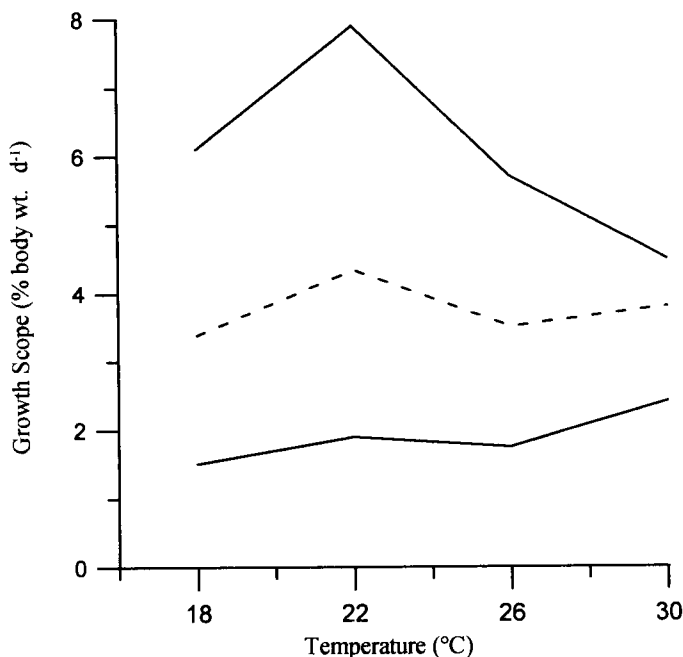


Figure 3. Growth scope for a 114 g smallmouth bass over the temperature range of 18-30 °C. The top line represents maximum daily consumption, the bottom line represents specific daily metabolic demand, and the dashed line represents the sum of energy losses due to metabolism, egestion, excretion, and specific dynamic action. The difference between maximum daily consumption and the dashed line represents growth scope at a given temperature.

Relations between growth scope (the difference between $C_{\max}$ and MET; Warren and Davis 1967) and temperature for a 114 g smallmouth bass also followed the pattern typically observed for fishes, with growth scope initially increasing with increasing temperature, peaking at some intermediate temperature (22 °C), and then declining as temperatures rise above the temperature at which $C_{\max}$ peaks (Jobling 1997). Although the upper thermal limit for growth (the upper temperature at which $C_{\max}$ and energy costs are equal) for a 114 g smallmouth bass was not reached by 30 °C, severe restriction in growth scope at that temperature (20% of maximum) and the likelihood of a continued decline in $C_{\max}$ and concomitant increase in energy costs at temperatures > 30 °C suggest that the upper thermal growth limit for sub-adult and adult fish would likely be near 31 °C. When water temperatures exceed 22 °C, and especially at temperatures $\geq$ 30 °C, restrictions in growth scope for sub-adult and adult smallmouth bass are expected to lead to diminished fish condition, size, and ultimately, fecundity (Bagenal 1967). For YOY smallmouth bass, however, growth scope does not begin to diminish with increasing temperature until 29 °C and remains high even at 30 °C (Post 1990, Hewett and Johnson 1992), so temperatures in this range would not be expected to have negative impacts on growth and condition of YOY smallmouth bass. In a laboratory study using constant temperatures ranging from 16 to 35 °C and ad libitum feeding, instantaneous growth rates of young juvenile smallmouth bass (mean wt. 29.4 g) were highest at 26 °C (Horning and Pearson 1973), intermediate in relation to optimum growth temperatures for YOY and adult fish.

Activity is a potentially important energy cost to smallmouth bass not included in relations between growth scope and temperature described here. Measurements of $C_{\max}$ and MET were conducted in the laboratory using fish held in enclosures that limited fish movement. In natural settings, energy loss due to activity can be highly variable and difficult to predict (Boisclair and Leggett 1989) but represents a potentially significant energy cost to fishes. Activity costs would reduce growth scope below levels presented in this study and could lead to negative growth potential for adult smallmouth bass, particularly at temperatures substantially greater than 22 °C.

Temperature fluctuation represents another factor which could potentially affect fish growth scope that was not considered in this study. Little is known of the effect of fluctuating temperatures on fish bioenergetics, although it has been suggested that fluctuating temperatures may reduce growth at temperatures above optimum, have little effect on growth at temperatures near optimum, and increase growth at temperatures below optimum (Jobling 1997). If so, then water temperatures fluctuating around a mean daily temperature substantially above 22 °C would be less conducive to growth of adult smallmouth bass than predicted from the constant temperatures modeled in this study. Since many aquatic environments, particularly small streams, can exhibit large daily fluctuations in temperature, studies investigating the effects of fluctuating water temperatures on fish energetics represent an important area for future research.

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